



DEALER BEST PRACTICE SERIES

Managing Payload on 992G Wheel Loaders

Application **Maintenance** **Site Management** **Component Rebuild** **Safety** **MARC Management**

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1.0 Introduction

This Best Practice illustrates the use of VIMS™ data for managing payloads in an application where 992G Wheel Loaders were required to deliver precise payloads.

Application background:

- Iron Ore loaded onto train cars for transport via public rail network
- Mining contractor responsible for loading the train cars.

This practice was developed when the contractor approached the Caterpillar Dealer to find a payload system that would address the following:

- Rail wagon axle loadings are critical to the public railway.
- Maximum payload critical for client.
- Validation that wagons loaded correctly by the Contractor



992G loading ore wagons

Precision payloads would rely upon VIMS functioning and reporting correctly, subsequently leading to:

- Lower cost per ton, by delivering maximum payloads to the Port.
- Eliminate overloading of the wagon /axles.
- Correct payload in wagons to maximize revenue for customer

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2.0 Best Practice Description

VIMS would be utilized for acquiring key payload data, as it was already on the 992G Wheel Loaders. Although, originally it did not report total truck weight, so Caterpillar changed the VIMS report and machine configuration file to enable this feature.

1. Every train loaded has an ID number (i.e. “7625”) and every wagon used to transport the ore, displays a “wagon ID” which was painted on the wagons by the railway company.
 - The Wagon ID numbering system has been categorized by target payload (for example: #100 to #200 would have a target weight of 68 tons; and #201 - #300 would have a target weight of 66 tons).



Wagon identification numbers painted to the side of ore wagons.

2. The train number is entered when the train stops. This number is entered into the “BLAST NUMBER” field.
3. The 992G operator begins to load the train, and after the first bucket, is prompted to enter the wagon ID (which is painted on the side of the ore wagon).



Prompt for Train Number

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4. VIMS prompts the operator to select the Material ID for loading “fines” or “lump” onto the train.
5. The operator will load two full buckets, take a payload reading, and adjust the third and final load accordingly, to match the target payload of the wagon.



Prompt for material ID

6. After the shift is complete and the machine is parked, the payload data is uploaded via wireless link directly into a program (provided by the dealer) on an office computer. Summary reports are generated to analyze payloads by train ID.
7. The total payload (tons) loaded, is verified by the Client when unloading at the port. Both the VIMS report and Client report are then reconciled. If excessive variation is discovered in each wagon, the operator will complete additional training on the payload system.

The VIMS payload reporting system is accurate, and a report confirms the payload weight variance on every wagon. (See next page).

<i>Train ID</i>	<i>Loader</i>	<i>Time</i>	<i>WagonID</i>	<i>Operator</i>	<i>Weight</i>	<i>Target</i>	<i>U/O target</i>
7625	7HR00213	1:22 AM	958	23	66.9	68	-1.1
7625	7HR00213	1:24 AM	932	23	69.3	68	1.3
7625	7HR00213	1:26 AM	938	23	69.4	68	1.4
7625	7HR00213	1:28 AM	953	23	68.4	68	0.4
7625	7HR00213	1:31 AM	964	23	67.8	68	-0.2
7625	7HR00213	1:33 AM	941	23	67.1	68	-0.9
7625	7HR00213	1:36 AM	924	23	68.4	68	0.4
7625	7HR00213	1:41 AM	908	23	68.7	68	0.7
7625	7HR00213	1:43 AM	909	23	69.3	68	1.3
7625	7HR00213	1:46 AM	926	23	68.6	68	0.6
7625	7HR00213	1:48 AM	935	23	68.0	68	0.0
7625	7HR00213	1:50 AM	942	23	67.5	68	-0.5
7625	7HR00213	1:52 AM	927	23	69.1	68	1.1
7625	7HR00213	1:55 AM	914	23	68.4	68	0.4
7625	7HR00213	1:58 AM	411	23	66.0	68	-2.0
7625	7HR00213	2:04 AM	768	23	71.9	73	-1.1
7625	7HR00213	2:06 AM	690	23	66.7	73	-6.3
7625	7HR00213	2:08 AM	657	23	72.7	73	-0.3
7625	7HR00213	2:11 AM	802	23	72.8	73	-0.2
7625	7HR00213	2:14 AM	626	23	73.0	73	0.0
7625	7HR00213	2:17 AM	780	23	72.6	73	-0.4
7625	7HR00213	2:20 AM	552	23	66.8	68	-1.2
7625	7HR00213	2:25 AM	790	23	73.0	73	0.0

992G VIMS Payload Data

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Train Payload Analysis Detailed Report Train ID: 7625

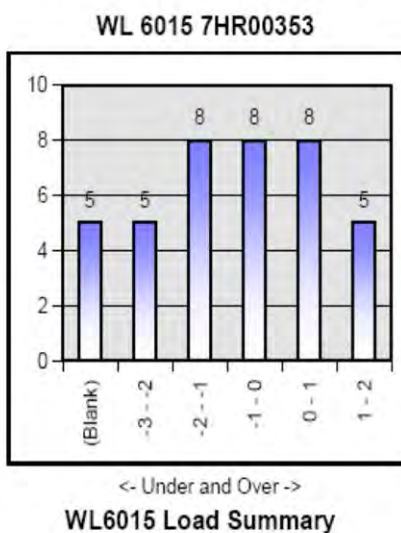
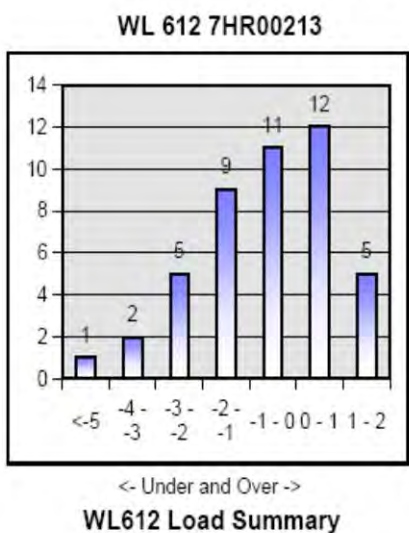
Total Loaded Weight: 5831

First Load: 27/09/2006 1:15:42 AM

Last Load: 27/09/2006 3:33:32 AM

Report Date: 10/10/2006

Count of Carriage Loads Under and Over Target and by How Much



WL 614 7HR00222

No Data for this machine

<- Under and Over ->

WL614 Load Summary

992G VIMS Data uploaded into Payload Analysis Detailed Report program

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3.0 Implementation Steps

1. Client paints the wagon ID numbers onto the ore wagons.
2. Dealer relocates keypad to enable easy operator input into VIMSTM.
3. Dealer creates Train Loading Report Program to verify actual wagon payloads.
4. Caterpillar had to change the report files and configuration files in the 992G loader system
5. Contractor has a master test weight fabricated and certified.
6. Dealer and contractor set-up VIMSTM. and trained 992G operators.



VIMS keypad relocated for easier operator access

4.0 Benefits

1. Train weight-to-Port weight is verified and accurate.
2. The axle loadings are checked and wagon loads verified.
3. The Customer has maximum payload railed to the Port. (One-half a ton short/ wagon could result in a potential revenue loss of \$7 million)
4. Correct axle weights reduce risk to the customer (in the case of a derailment).
5. Correct payload in wagons maximizes revenue for customer.

5.0 Resources Required

1. Client paints wagon ID numbers to wagons	- \$5,000
2. Dealer made report program. (No charge)	- \$ 0
3. Contractor fabricates calibration weight & validates.	- \$7,000
4. Dealer relocates VIMS key pad	- \$1,000
5. Dealer conducts site training. (No charge)	- \$ 0
TOTAL	- <u>\$13,000</u>

6.0 Supporting Attachments / References

None applicable.

7.0 Related Best Practices

None applicable.

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8.0 Acknowledgements

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